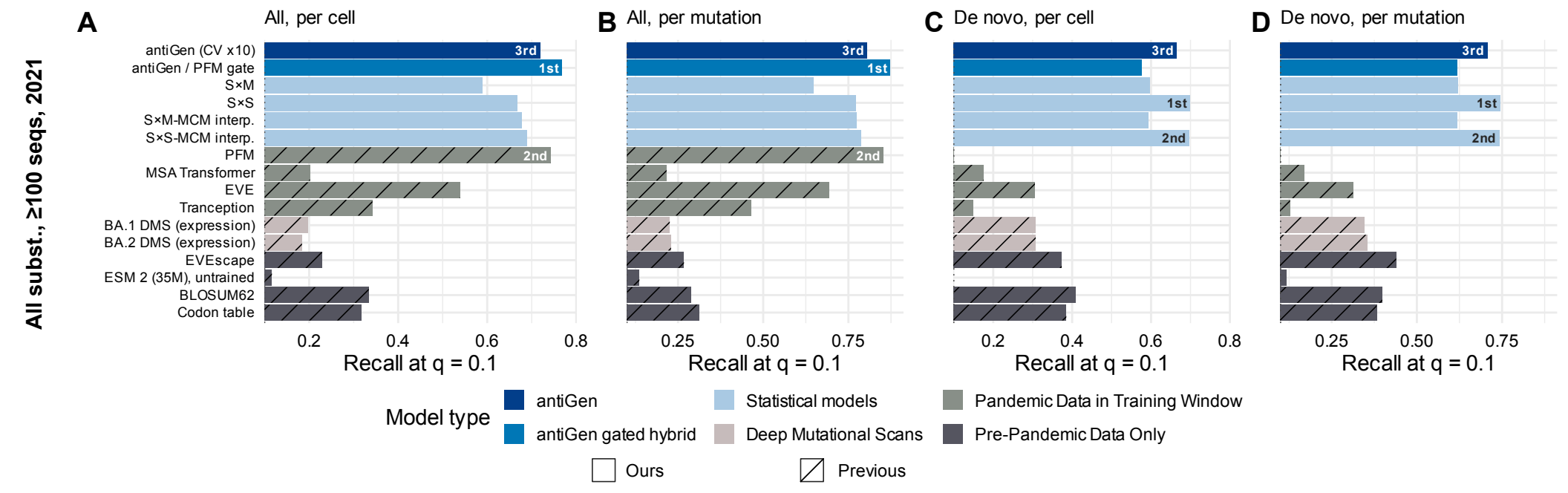
All subst., ≥100 seqs, 2021
A
All, per cell
B
All, per mutation
C
De novo, per cell
D
De novo, per mutation
antiGen (CV x10)
antiGen / PFM gate
S×M
S×S
S×M-MCM interp.
S×S-MCM interp.
PFM
MSA Transformer
EVE
Tranception
BA.1 DMS (expression)
BA.2 DMS (expression)
EVEscape
ESM 2 (35M), untrained
BLOSUM62
Codon table
3rd
1st
2nd
0.2
0.4
0.6
0.8
0.25
0.50
0.75
Recall at q = 0.1
Model type
antiGen
antiGen gated hybrid
Statistical models
Deep Mutational Scans
Pandemic Data in Training Window
Pre-Pandemic Data Only
Ours
Previous